

CSCE 633 MACHINE LEARNING · MODULE 03 · LECTURE NOTES

Bias, Variance, and Capacity

Why more model is not more better.

MODULE OUTCOMES

- State and interpret the bias–variance decomposition.
- Read a learning curve and diagnose from it.
- Explain capacity and how it is controlled.
- Distinguish underfitting from overfitting by evidence.
- Explain double descent and what it does not overturn.

1 The decomposition

$$E[(y - f_{\square}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

For squared loss, the expected error over random draws of the training set decomposes exactly into **three terms**. **Bias** is how wrong the *average* fitted model is — error from the model class being too restricted to represent the truth. **Variance** is how much the fitted model moves as the training sample changes — error from fitting this particular sample rather than the underlying pattern. σ^2 is **irreducible noise** in the labels themselves, and it is the floor: no model, however good, goes below it. **You control two of the three, and they trade against each other**, which is why the question is never 'is this model good' but 'which term is limiting me'.

Symptom	Diagnosis	What helps
Training error high, validation error close to it	High bias (underfitting).	More capacity, better features, or less regularisation. More data will not help — see the callout.
Training error low, validation error much higher	High variance (overfitting).	More data, stronger regularisation, or less capacity. Also data augmentation, which raises the effective sample size.
Both errors high <i>and</i> the gap is large	Both at once, which is common on hard problems.	Features first — they attack bias without increasing variance — then capacity.
Both errors low	Done.	Stop, and then be suspicious. This is the signature of leakage as often as of success (Module 12 §1), and the right response to an unexpectedly good result is an audit rather than a celebration.

Symptom	Diagnosis	What helps
Validation error rising as training continues	Overfitting over training time rather than over capacity.	Early stopping , which is capacity control in disguise (§3).

MORE DATA FIXES VARIANCE AND DOES NOTHING FOR BIAS

High variance means the model is fitting the particular sample. More samples make that sample a better approximation of the distribution, so the error from fitting it shrinks — directly and reliably.

High bias means the model class cannot represent the truth at all. A straight line fitted to a parabola is exactly as wrong with a million points as with a hundred; the extra data simply pins down the best wrong answer more precisely.

So diagnose before collecting. Data acquisition is usually the most expensive action available — in money, in time, and frequently in labelling effort — **and it is frequently the wrong one**, chosen because it is the obvious response to 'the model is not good enough'.

The learning curve answers this directly (§2). If the validation curve is still descending at your current sample size, more data will help and you can extrapolate roughly how much. **If the two curves have converged, more data buys nothing and the money should go elsewhere.** This is one of the highest-value diagnostics in applied machine learning and it costs a handful of training runs.

2 Learning curves

Plot TRAINING and VALIDATION error against TRAINING SET SIZE (not against epochs -- they answer different questions).

HIGH BIAS

```
err |
    |_____ val
    |_____ train
    |
    +----- n
```

Converged, HIGH.
More data will NOT help.
Add capacity or features.

HIGH VARIANCE

```
err |
    | \_____ val
    |
    | _____ train
    |
    +----- n
```

Large persistent GAP.
More data WILL help.
Or regularise.

READING THE SHAPES

converged and low	-> done (and be suspicious: Module 12)
converged and high	-> bias; add capacity
gap still closing	-> get more data; it is working
gap flat and wide	-> regularise; data alone is not enough
validation RISING	-> early stop

Plotting against dataset size and plotting against epochs answer different questions, and the second is routinely mistaken for the first. The epoch curve tells you when to stop training this model; **the data-size**

curve tells you whether to collect more data, which is a far more consequential decision. **This single plot answers 'what should I do next?' better than any other diagnostic available**, and it costs one training run per point — which, for the models in this course, is minutes.

3 Capacity

- **Parameter count.** The obvious measure, and **the least reliable predictor of generalisation** — which §4 makes vivid.
- **Function class.** Polynomial degree, tree depth, number of neighbours, kernel choice. Usually the most interpretable knob.
- **Regularisation strength** (Module 04), which reduces *effective* capacity while leaving the parameter count untouched — the model can still express complex functions and is penalised for doing so.
- **Training time. Early stopping is capacity control**, because an under-trained model has not yet reached the flexible region of its function class. This is not an analogy; for linear models with gradient descent it can be shown equivalent to a form of ridge regression.
- **And data augmentation**, which raises the effective sample size rather than lowering capacity — a different mechanism with the same effect on the generalisation gap. **'Effective capacity' is the useful notion and it is not the parameter count**, which is exactly what makes double descent possible.

4 Double descent

THE CURVE HAS A SECOND DESCENT

The classical picture: as capacity grows, test error falls to a minimum and then rises, because variance overtakes the reduction in bias. A U-shaped curve, and the job is to find its bottom.

What is observed in very large models: test error rises to a peak at the *interpolation threshold* — the capacity at which the model can just exactly fit the training data — **and then falls again, sometimes below the classical minimum**, as capacity continues to grow.

So heavily overparameterised models can generalise well, which the classical account did not merely fail to predict but positively argued against. This is the empirical fact underneath modern deep learning working at all.

It does not overturn the bias–variance decomposition. That decomposition is an algebraic identity — it is always true, for any estimator. **What was overturned is the separate assumption that variance necessarily increases with parameter count.** Among the many functions that interpolate the training data, gradient descent appears to find low-complexity ones — implicit regularisation from the optimiser itself — so the effective capacity does not grow the way the parameter count does.

WHAT THIS MEANS FOR YOUR PROJECT

At the scale of this course, the classical picture holds. With a few thousand samples and a tabular problem, adding capacity past the sweet spot degrades test performance exactly as the U-curve predicts, and you should act accordingly.

Double descent appears with very large models on very large data — CSCE 636's territory, not this module's. Knowing it exists prevents you from concluding that a large model is *necessarily* overfitting, which is a conclusion people reach from the textbook curve alone.

And the diagnosis procedure is entirely unchanged. Plot the learning curves, read the gap, act on it. **The empirical method survives the theory changing**, which is precisely why the course emphasises it.

The genuinely useful lesson here is epistemic. A well-established picture, taught confidently for decades, turned out to be incomplete — and the correction came from measurement rather than from theory, when people trained models larger than the theory contemplated and plotted what happened. **That is a reason to keep plotting rather than reasoning from the remembered curve**, and it generalises well beyond this subject.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
James et al. — An Introduction to Statistical Learning, chapter 2 (free PDF)	The decomposition of §1 , derived, with the clearest available figures.
Andrew Ng — Machine Learning Yearning (free)	The §2 diagnostic procedure as an engineering discipline — short chapters, entirely about deciding what to do next.
Belkin, Hsu, Ma & Mandal — Reconciling modern machine learning practice and the bias-variance trade-off (free)	The double descent paper . The figures are the argument.
Hastie et al. — The Elements of Statistical Learning, chapter 7 (free PDF)	Model assessment and selection at depth, including the effective degrees of freedom notion behind §3.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Fit polynomials of degree 1 to 15** to noisy data from a known function. Plot training and test error against degree.

- 2 **Estimate bias and variance empirically** by refitting on many bootstrap samples and measuring the spread of predictions.
- 3 Plot learning curves for your project dataset and **write the diagnosis on the plot**.
- 4 **Demonstrate that more data does not fix bias**: fit a line to a parabola at $n = 100$ and $n = 100,000$.
- 5 Demonstrate that more data does fix variance, with the same model on the same sizes.
- 6 Plot error against epochs and against dataset size for the same model. Explain what each tells you.
- 7 **Show that early stopping acts as regularisation** by comparing it against an explicit penalty.
- 8 Push a model past the interpolation threshold on a small dataset and plot test error against capacity. **Report whether you see a second descent**.
- 9 Take an unexpectedly good result and audit it for leakage before believing it.
- 10 For your project, state which term currently limits you and what you will do about it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the decomposition and say which terms you control.
- 2 Give five learning-curve symptoms and the diagnosis for each.
- 3 Why does more data fix variance and not bias?
- 4 Why plot against dataset size rather than epochs?
- 5 Name five things that control effective capacity.
- 6 Why is early stopping a form of capacity control?
- 7 State double descent and say precisely what it does and does not overturn.
- 8 What should you do when both errors are low?